# Auto-Updating Intrusion Detection System for Vehicular Network: A Deep Learning Approach Based on Cloud-Edge-Vehicle Collaboration

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***Abstract*—Intrusion detection systems play a crucial role in ensuring the safety of vehicle driving. Traditional intrusion detection systems face challenges in efficiently extracting key features from network traffic, resulting in inefficiency. To obtain better detection performance, many researchers adopted deep learning to build intrusion detection systems for vehicular network. However, these systems can only identify learned network attacks, while failing to identify the unknown attacks present in open scenarios. In this paper, we propose a deep learning-based auto-updating intrusion detection system for vehicular network, which uses a two-layer filter to effectively sense unknown attacks in open scenarios and proactively updates the attack recognizer to enable it with the ability to identify unknown attacks. In addition, we design a cloud-edge-vehicle collaborative model updating approach, where the updating task of the attack recognizer is deployed on a cloud server to reduce the computational load of the vehicle. The comprehensive experiments on three open scenarios show that the updated attack recognizer can accurately identify both known and unknown attacks.**

***Index Terms*—Vehicular network, intrusion detection system, deep learning, unknown attack.**

## I. INTRODUCTION

WITH the application of sensor and wireless communication technologies in transportation systems, the world has transitioned into an era of intelligent transportation [1]. Smart vehicles are equipped with an increasing number of safety and entertainment applications, such as road prediction, assisted driving, and audio/video streaming downloads [2]. However, while vehicle users enjoy the services of vehicular applications, malicious attackers may launch network attacks on vehicles and cause serious traffic accidents [3], [4], [5]. In particular, vehicular networks operate in open scenarios, and malicious attackers will constantly create new types of attacks to attack vehicular networks. Since this new type of attack is unknown to the intrusion detection system, it is difficult for the detection system to recognize the new unknown attack accurately, and the vehicle is more vulnerable to the attack. Therefore, to ensure the security of vehicular network, the development of an efficient and reliable intrusion detection system for vehicular network becomes increasingly important [6]. Traditional intrusion detection methods mostly use fixed rules and features as the basis for network attack detection [7], [8]. However, in large-scale and complex vehicular network environments, these methods prove ineffective as extracting useful features from network flow becomes challenging [9], [10]. In recent years, the application of deep learning in image recognition and natural language processing has led researchers to explore its potential in constructing intrusion detection systems for vehicular network, yielding improved results [11]. Despite these advancements, most existing vehicular network intrusion detection methods can only identify learned network attacks and cannot detect possible unknown attacks [12]. Since the types and number of network attacks are not constant in the real world, there will always be unknown attacks that have not been learned by the intrusion detection system. As shown in Fig. 1, a malicious attacker launches three types of network attacks on a target vehicle. Since the intrusion

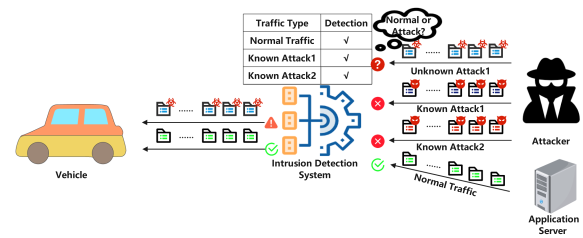


Fig. 1. The process of recognizing attacks by intrusion detection system.

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detection system has learned the data features of two types of known attacks, the network flows of the known attacks are easily recognized and defended by the detection system. However, the third type of attack is an unknown attack that has not been learned by the intrusion detection system, which makes it difficult for the detection system to recognize this attack.

In particular, many researchers utilize datasets consisting of fixed types of attacks to train intrusion detection systems, leading to the system's outputs being confined within a limited range, while unknown attacks do not fall within this range. Therefore, when faced with an unknown attack, the intrusion detection system produces two possible results [13]. The first result is that the unknown attack is judged as normal traffic, resulting in the vehicle being victimized directly. The second result is that the unknown attack is judged to be a known attack, leading the vehicle to deal with the unknown attack using methods that defend against known attacks. However, the defenses employed by the vehicle may be ineffective due to the different characteristics of the different attacks [14].

To address the above problem, we propose an auto-updating intrusion detection system for vehicular network, which periodically updates the attack recognizer to identify known/unknown network attacks present in open scenarios. The basic idea of the proposed system is mainly 1) filtering out the network flows of unknown attacks from the mixed traffic, 2) since there may be more than one kind of unknown attack, the obtained unknown attack flows need to be classified, 3) based on the network flows of unknown attacks and the initial dataset, a new dataset containing normal flows, known/unknown attacks is formed and the attack recognizer is updated using this dataset. 4) the updated attack recognizer has the ability to recognize known/unknown attacks. Moreover, it is important to note that the update of the attack recognizer cannot be performed on the vehicle due to its limited computing and storage capacity. As a result, we design a cloud-edge-vehicle cooperative model update method to avoid the vehicle computation task overload problem. In summary, the contributions of this paper are summarized as follows:

- We propose a deep learning-based auto-updating intrusion detection system for vehicular network, which can effectively detect unknown attack flows present in open scenarios and proactively update the attack recognizer to improve the ability to recognize unknown attacks.
- We propose a cloud-edge-vehicle collaboration-based model updating approach, which deploys the attack recognizer updating task on cloud servers to reduce the computational and storage load of vehicles.
- The comprehensive experiments on three open scenarios show that the proposed intrusion detection system for vehicular network can effectively identify both known and unknown attacks, and outperform the state-of-the-art methods in terms of precision, recall and F1-score.

The remainder of this paper is organized as follows. The related work is described in Section II. The preliminaries are described in Section III. In Section IV, we introduce the proposed auto-updating intrusion detection system for vehicular network. In Section V, we evaluate the performance of the proposed system. Finally, we summarize our work in Section VI.

## II. Related Work

Recently, there has been a proliferation of research on intrusion detection systems. In this section, we will discuss four main categories of intrusion detection systems.

### A. Anomaly-Based Intrusion Detection Methods

Anomaly-based intrusion detection system is a popular method for detecting unknown attacks, which utilizes the features of normal traffic to build an attack recognizer for detecting the presence of anomalous attack traffic in the network. Zhang et al. [15] proposed an anomaly detection method for unknown attacks, which uses Kmeans and isolated forests to detect the presence of anomalous traffic in mixed traffic. To avoid botnet attacks on network devices, Sriram et al. in [16] proposed a botnet attack detection framework based on a multilayer perception, which identifies malicious traffic based on statistical features of network flows. Considering that the training data may contain privacy information of users, Shi et al. [17] proposed a hybrid intrusion detection system based on differential privacy and machine learning, which uses one class support vector machine and local outlier factor algorithm for network traffic detection. To reduce the false alarm rate of network anomalies, Li et al. [18] propose an anomalous traffic detection method based on a bidirectional gated recurrent unit and an attention mechanism, which uses a stacked sparse auto-encoder to extract useful features of network traffic, so as to improve the accuracy of detection. The above-mentioned anomaly-based intrusion detection methods all achieve good results in terms of detection accuracy. However, these methods can only determine whether the network contains anomalous traffic, and cannot construct suitable defenses for such anomalous traffic.

### B. Machine Learning-Based Intrusion Detection Methods

To formulate appropriate defense strategies against different types of network attacks, many researchers have conducted fine-grained analyses for network anomalies (i.e., categorizing anomalous network traffic). In this subsection, we focus on machine learning-based intrusion detection methods, while in the next subsection, we will discuss deep learning-based intrusion detection methods. Alani et al. [19] propose a two-layer intrusion detection method for malicious attacks in networks, which uses multiple machine learning algorithms for packet-level data and flow-level data. Yang et al. [20] proposed a multi-layer hybrid intrusion detection method that uses decision trees, random forests, and other algorithms as base classifiers to identify both inside and outside vehicle anomalous messages. To avoid distributed denial of service attacks on vehicles, Anyanwu et al. [21] proposed a vehicular network intrusion detection method based on the RBF-SVM kernel, which uses a grid search cross-validation algorithm to select the hyperparameters of the model. Jin et al. [22] proposed an intrusion detection method based on outlier detection, oversampling and metric learning for achieving fast detection of malicious attacks in a vehicular network environment. These aforementioned schemes achieve the fine-grained classification of network attacks. However, the accuracy of machine learning algorithms is relatively low due to their weak learning ability on large-scale datasets. Moreover, machine learning algorithms rely on manual feature selection, which increases the training cost.

### *C. Deep Learning-Based Intrusion Detection Methods*

To improve the accuracy of identifying network attacks at a fine-grained level, more and more researchers have used deep learning to train intrusion detection models. Deep learning can use multilayer neural networks to perform automatic feature extraction on large-scale data and obtain high accuracy rates. To detect false message attacks in vehicular networks, Yu et al. [23] proposed an intrusion detection system based on long and short-term memory networks, which uses vehicle message sequences as input to determine emergency message classes. Shu et al. [24] proposed a GAN distributed training-based intrusion detection method for the biased flow problem in vehicular networks, which increases the robustness of the model by collecting network traffic from multiple regions. Tian et al. [25] proposed an intrusion detection method based on residual convolutional neural networks, which uses a distributed deep learning system to improve the accuracy of the classification model. These intrusion detection schemes mentioned above can achieve accurate and fine-grained identification of network attacks. However, since the models of these solutions usually do not have the ability to sense unknown attacks, they cannot effectively identify unknown attacks in open scenarios.

### *D. Physical Signal-Based Intrusion Detection Methods*

To realize timely detection of network attacks, many researchers have proposed physical signal-based detection methods for vehicular network attacks. Deng et al. [26] proposed an in-vehicle intrusion detection system called IdentifierIDS, which detects both periodic and non-periodic malicious frames by establishing a voltage fingerprint for each CAN identifier (ID) and does not consume the limited bandwidth of the CAN bus. It is worth noting that IdentifierIDS does not rely on ECU and ID mapping knowledge, which greatly enhances utility. Qin et al. [27] proposed a collaborative cloud-vehicle intrusion detection system based on multidimensional features, which solves the data heterogeneity problem by abstracting different vehicle data into the same feature space. Comprehensive experiments on three real vehicles show that the scheme can effectively detect potential attacks and greatly improve the robustness of the detection system. To reduce the consumption of CAN bus bandwidth resources by the intrusion detection system, Xun et al. [28] proposed an intrusion detection system based on vehicle voltage signals, which utilizes the side channel information of the vehicle ECU to detect malicious messages and attack sources. It is worth mentioning that the experiments of this system on real vehicles have high detection accuracy and it takes only 0.537 ms to detect a data frame, which means that the system can accurately recognize malicious attacks in a shorter time. To detect malicious spoofing attacks on the CAN bus, Levy et al. [29] proposed a spoofing attack detection scheme based on voltage signals, which utilizes the physical side channel to detect and localize malicious attacks and can adaptively adjust the detection model according to the surrounding environment.

In summary, many researchers have proposed a large number of intrusion detection schemes to secure the network of vehicles. However, most vehicular intrusion detection systems rarely consider the detection of new unknown attacks, resulting in vehicles being vulnerable to potential attack threats. In this paper, we propose an auto-updating vehicular intrusion detection system based on cloud-edge-vehicle collaboration. The proposed scheme can effectively sense unknown attacks present in open vehicular network scenarios and improve the detection capability by continuously updating the attack recognizer. In addition, the updating process of the attack recognizer is completed by a cloud server, which greatly reduces the computation and storage overhead of the vehicle.

## III. Preliminaries

In this section, we formulate the problem of identifying network attacks in open scenarios and then present the main components of the proposed attack recognizer.

### *A. Problem Formulation*

Let $D^t = \{(x_1, y_1), (x_2, y_2), \ldots, (x_i, y_i), \ldots, (x_n, y_n)\}$ be a labeled training dataset that contains $M$ known network attacks, where $t$ denotes the current time. Define $x_i$ as the $i$-th sample in dataset $D^t$ and $y_i$ as the label corresponding to $x_i$. Define the current known attack set as $C = \{c_1, c_2, \ldots, c_j, \ldots, c_M\}$ and $c_j$ corresponds to the $j$-th known network attack. Using dataset $D^t$ to train an initial attack recognizer $F^t$ that can identify the known $M$ network attacks. Assume that at time $t+1$, the attacker launches a novel network attack $c_{M+1}$ and the existing attack recognizer $F^t$ cannot effectively identify the unknown attack $c_{M+1}$. To prevent the vehicular network against attackers, we need to update the attack recognizer $F^t \to F^{t+1}$ to effectively identify the $M+1$ attacks present in the current scenario. The update of recognizer $F^t \to F^{t+1}$ relies on the new dataset $D^{t+1}$, which contains samples with unknown attacks and the corresponding labels. Therefore, the core challenge of constructing an intrusion detection system for vehicular network in open scenarios is to efficiently detect unknown attack flows and form a dataset containing unknown attacks. The identification of network attacks in the open scenario can be divided into four subproblems as follows:

1) To identify existing known attacks, we need to train a known attack recognizer with high accuracy.
2) Design an unknown attack filter to identify whether an attack is known or unknown.
3) To determine how many kinds of unknown attacks exist in the open scenario, an efficient unknown attack classifier needs to be designed.
4) Generate a labeled dataset $D^{t+1}$ containing known attack flows and unknown attack flows based on the unknown attack filter and unknown attack classifier, and use $D^{t+1}$ to update the attack recognizer.

### B. Composition of Attack Recognizer

The proposed attack recognizer mainly consists of convolutional neural network (CNN) and self-attention mechanism. The CNN is used to extract useful information about the network flow, and the self-attention mechanism correlates these information with each other to form a global information representation [30], [31], [32]. It is worth mentioning that CNNs are further divided into 1D-CNN, 2D-CNN, and 3D-CNN to handle data of different dimensions (for example, 1D-CNN is commonly used to handle text data). Since network flow is a kind of text data, we use 1D-CNN and self-attention mechanism to build an attack recognizer.

*1) 1D-CNN:* A classical 1D-CNN model usually consists of multiple convolutional layers and pooling layers. The convolutional layer is mainly used to extract the features of the network flow. The specific convolution process is as follows:

$$h_i = \sigma\left(\sum_{j=0}^{m-1} w_j x_{i+j} + b\right) \tag{1}$$

where $\sigma$ is the activation function, $m$ is the size of the convolution kernel, $w_j$ is the $j$-th weight of the convolution kernel, $b$ is the offset, $x_{i+j}$ is the $(i+j)$-th element of the input sequence, and $h_i$ is the output value of the convolution operation. To obtain better performance, we use multi-layer 1D-CNN for feature extraction of network traffic. The final convolution result $C^{res}$ needs to be input to the pooling layer for dimensionality reduction to reduce the risk of model overfitting and increase the training speed. The specific pooling process is as follows:

$$Z = MaxPooling(C^{res}) \tag{2}$$

*2) Self-Attention Mechanism:* Generally, the self-attention mechanism uses three matrixes of "query", "key" and "value" to construct the relational features between vectors. For convenience, "query", "key" and "value" are denoted as $Q$, $K$ and $V$, respectively, and their values are calculated as follows:

$$\begin{cases} Q &= Z \cdot W^Q \\ K &= Z \cdot W^K \\ V &= Z \cdot W^V \end{cases} \tag{3}$$

where $W^Q$, $W^K$, and $W^V$ are learnable weight matrices.

According to the matrix $Q$ and the matrix $K$, the value of the attention matrix $A$ can be calculated by (4)

$$A = Softmax\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) \tag{4}$$

where the $Softmax$ is a activation function, and $d_k$ is the dimension of $K$. Note that the matrix $A$ contains the relationship between any two elements of the pooling result $Z$. Then, we use the attention matrix $A$ to extract the key information from the matrix $V$ and generate a sequence $O'$ that can reflect the relationship between the elements. The value of $O'$ can be calculated by the (5)

$$O' = Attention(Q, K, V) = A \cdot V \tag{5}$$

Note that the $O'$ is the output of a single-head attention mechanism, and single-head attention suffers from the drawback of over-focusing attention on a certain location. Therefore, we avoid the over-focusing problem by training multi-groups of different parameter matrices to generate a series of $O'$. Then, the output sequence $O$ of the multi-headed attention mechanism can be calculated by (6)

$$O = (O'_1 \oplus O'_2 \oplus \cdots O'_H) \cdot W^O \tag{6}$$

where $W^O$ is a learnable weight matrix, $H$ is the number of heads of the attention mechanism.

## IV. Auto-Updating Intrusion Detection System for Vehicular Network

The proposed auto-updating intrusion detection system for vehicular network consists of four components: attack recognizer, unknown attack filter, unknown attack classifier, and update attack recognizer. The overview of the system is shown in Fig. 2. Note that the four parts of the proposed system correspond to each of the four subproblems in the problem formulation.

### A. Attack Recognizer

To enable the defense against network attacks, we need to design attack recognizer with high accuracy to identify the attacks present in the vehicular network. As declared in subsection B of the preliminaries, the attack classifier mainly consists of a 1D-CNN and a multi-headed attention mechanism. Initially, the attack recognizer extracts the key features of the network flow using 1D convolution and uses maximum pooling to reduce the risk of model overfitting. Subsequently, the pooled feature vectors are fed to the multi-headed attention mechanism to capture the correlation between features. Then, we obtain a hidden vector $O$ containing the key information of the network flow. Finally, the vector $O$ is flattened and fed into a linear layer with a $Softmax$ activation function. Note that the $Softmax$ activation function is often used for classification tasks, and its output is the predicted probability of each category. The output of the linear layer and the Softmax activation function are shown in (7) and (8)

$$y = [y_1, y_2, \ldots, y_M, y_{M+1}] = (W^T \cdot O) + b^f \tag{7}$$

$$p_j = \frac{exp(y_j)}{\sum_{i=1}^{M+1} exp(y_i)} \tag{8}$$

where $W^T$ is the weight matrix, $b^f$ is the offset, $y$ is the output of linear layer, $p_j$ is the predicted probability of the $j$-th class, and $M+1$ denotes normal traffic and M types of known attacks. According to (8), we can obtain the probability $p = [p_1, p_2, \ldots, p_{M+1}]$ that a flow belongs to normal traffic and each of the known attacks, where $\sum_{i=1}^{M+1} p_i = 1$ and $p_i \leq 1$. The type of the network flow is determined by the maximum value in $p$, which is denoted as $p^* = \text{Max}(p)$.

To sum up, the attack recognizer has been built. However, the initial attack recognizer can only identify $M$ known attacks and cannot identify unknown attacks that may exist in open

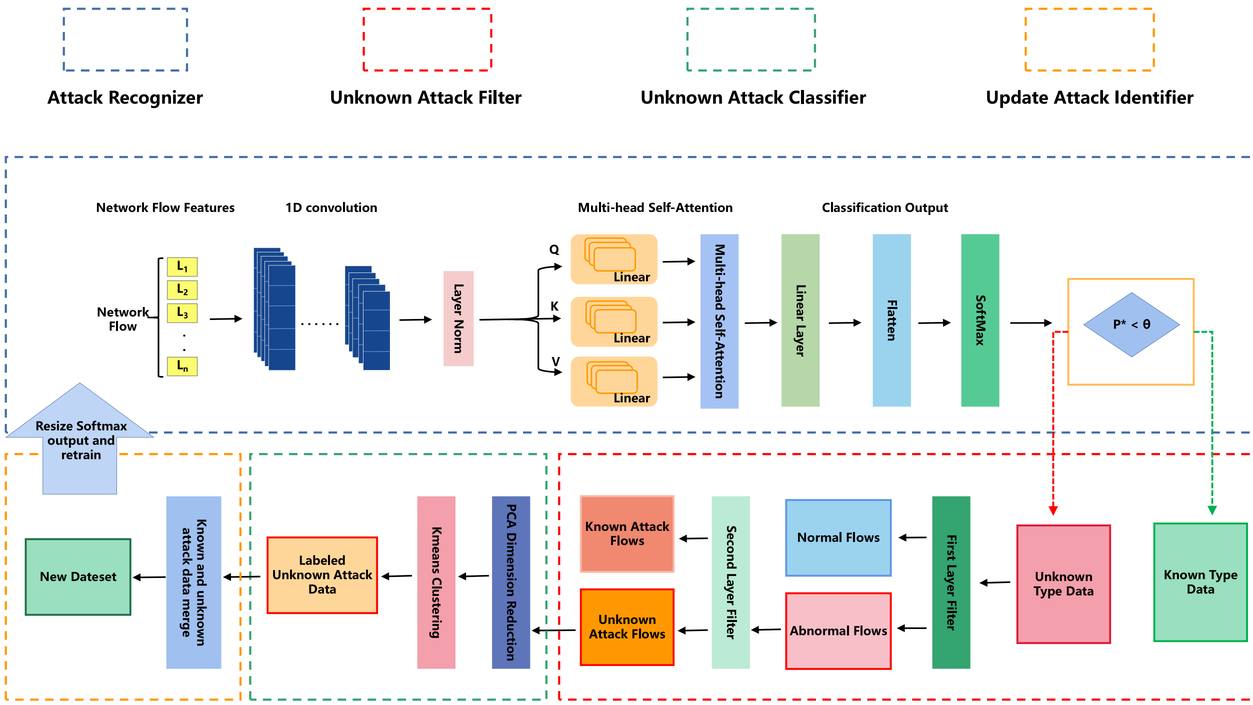


Fig. 2. Overview of auto-updating intrusion detection system for vehicular network.

scenarios. Inspired by Sun et al. [33] and our experimental validation, we find that unknown and known attacks show different distributions on the probability. More specifically, the $p^*$ value of a known attack is slightly less than 1, while the $p^*$ value of an unknown attack is much less than 1. Therefore, we define a threshold $\theta$ to determine whether a network flow belongs to a known class or not. When $p^* > \theta$, the flow is determined to be a known class, otherwise it belongs to an uncertain class (i.e., it may be either a known or an unknown class). We set the flows with $p^* > \theta$ as $x^k$, and the flows with $p^* \leq \theta$ as $x^u$.

### *B. Unknown Attack Filter*

To further determine whether a network flow is an unknown attack flow, we need to design a filter to eliminate the normal flows and known attack flows in $x^u$.

Let $D^t$ be the training dataset with labels at the initial time $t$, and $D^t$ contains both normal traffic samples and known attack samples. Since we do not know any useful information about the unknown attack at the initial time $t$, a discriminator for the unknown attack is difficult to construct. However, the dataset $D^t$ contains information about both normal flows and known attack flows. Therefore, a feasible approach is to build a two-layer filter to sequentially filter out the normal flows and known attack flows in $x^u$ based on the features of normal and known attack flows. Then, the remaining network flows are the unknown attack flows. Note that the first layer of the two-layer filter is used to identify whether a network flow is normal or not, while the second layer of the two-layer filter is used to detect unknown attack traffic. The reason for this is to avoid some normal flows being incorrectly recognized as attack flows.

**Algorithm 1:** Unknown Attack Filter.

**Input:** Dataset $D^t$ and Dataset $x^u$
**Output:** Unknown Attack Dataset $D_u$
Define $D_n^t$ as the normal flow in the dataset $D^t$;
Define $D_k^t$ as the known attack flows in the dataset $D^t$;
Building OCSVM-based first layer filters using $D_n^t$;
$\Phi^n = OCSVM(D_n^t)$;
Building OCSVM-based second layer filters using $D_k^t$;
$\Phi^k = OCSVM(D_k^t)$;
$\Phi^n$ and $\Phi^k$ are used to implement filtering of normal flows and known attack flows;
$x_1^u = \Phi^n(x^u)$;
$D_u = \Phi^k(x_1^u)$;
return $D_u$;

It is worth mentioning that the two-layer filter is built based on one-class support vector mechanism (OCSVM) [34], [35]. The OCSVM is a commonly used anomaly detection algorithm that builds a decision boundary in a high-dimensional space by features of training data. The decision boundary can cover most of the normal data and separate out the abnormal data. The objective function of the decision boundary is as follows:

$$\text{Min}_{w,\zeta_i,\rho}\left[\frac{1}{2}||w||^2 + \frac{1}{vn}\sum_{i=1}^{n}\zeta_i - \rho\right] \tag{9}$$

$$\text{s.t.} \begin{cases} (w \cdot \phi(x_i)) \geq \rho - \zeta_i \\ \zeta_i \geq 0 \end{cases}$$

where $w$ is the normal vector of the hyperplane, $v$ is a hyperparameter to adjust the slack degree, $n$ is the number of samples,

$\zeta_i$ is the slack variable, $\rho$ is the offset, and $\phi(x_i)$ is the radial basis kernel function. The detailed filter construction is shown in Algorithm 1, and based on the unknown attack filter, we can obtain $D_u$ containing the unknown attack flows.

### C. Unknown Attack Classifier

Although the dataset $D_u$ containing unknown attacks is obtained after the two-layer filter, we do not know how many unknown attacks are contained in $D_u$. Therefore, to further determine the number of unknown attack types in $D_u$, we need to perform clustering operation for $D_u$. The process of clustering is to divide $D_u$ into a number of clusters, and the network flows in each cluster have some similarity to each other. To improve the accuracy of the clustering results and reduce the computational complexity, we use principal component analysis (PCA) to perform dimensionality reduction for dataset $D_u$. PCA is a classical data dimensionality reduction algorithm that reduces the correlation between features by changing the coordinate system to find the main features in the data [36], [37]. Specifically, PCA first normalizes the data so that each feature with mean of 0 and variance of 1. The data normalization process is as follows:

$$T = \frac{q-\mu}{\tau} \tag{10}$$

where $q$ is the feature value, $\mu$ and $\tau$ are the mean and the standard deviation of the feature value set respectively, and $T$ represents the normalization result. Then, we calculate the covariance matrix to find a feature space that minimizes the correlation between the features. The covariance matrix is calculated as shown in (11)

$$CM = \frac{\sum_{i=1}^{G}(x_i - \overline{x})(x_i - \overline{x})^T}{G-1} \tag{11}$$

where $G$ is the number of samples, $x_i$ is the eigenvector of the $i$-th sample in $D_u$, and $\overline{x}$ is the mean vector of samples. According to the matrix $CM$, we can calculate its corresponding eigenvector $V$ and eigenvalue λ. The size of the eigenvalue reflects the importance of the feature vector. Therefore, we sort the eigenvector $V$ in descending order by eigenvalue λ and select the first k eigenvectors to realize data dimensionality reduction. The dataset after dimensionality reduction is denoted as follows:

$$D^{pca} = XV^k \tag{12}$$

where $X$ is the original data matrix and $V^k$ is a matrix consisting of the first k eigenvectors.

Subsequently, we use the Kmeans clustering algorithm to analyze how many unknown attacks are contained in $D^{pca}$. Kmeans is a distance-based clustering algorithm that divides a set of data into $\kappa$ classes by continuous iterations, where $\kappa$ is a hyperparameter. In the initial time, Kmeans selects $\kappa$ clustering centers. Then, the distance from sample point $s$ to each cluster center is calculated according to (13), and the minimum distance is chosen to determine the class to which the sample point belongs.

$$d_j = \sqrt{(s-c_j)^2} \tag{13}$$

**Algorithm 2:** Unknown Attack Classification.

**Input:** Dataset $D^{pca}$ and $\kappa_{max}$
**Output:** Multiple unknown attack data clusters $C^u = \{d_1, d_2, \cdots, d_\kappa\}$

Score_list = [];
**for** $\kappa \leftarrow 2$ **to** $\kappa_{max}$ **do**
  According to Equations (13) and (14) to cluster for dataset $D^{pca}$, and the clustering results are denoted as $C^u_\kappa = \{d_1, \cdots, d_\kappa\}$;
  Scoring of clustering result $C^u_\kappa$ by silhouette coefficients, and the score is marked as $S_\kappa$;
  Score_list ← $S_\kappa$
Select the maximum score $S^{max}$ from Score_list and mark its corresponding $\kappa$ as $S^{max}_{index}$;
// $\eta$ is the score threshold;
**if** $S^{max} > \eta$ **then**
  **if** $S^{max}_{index}\ != 2$ **then**
    Select the $C^u$ obtained when $\kappa$ is $S^{max}_{index}$;
    Return $C^u$;
  **else**
    Calculate the inter-cluster distance $Dis^{inter}$ of two clusters;
    Calculate the intra-cluster distances $Dis^{intra}_1$ and $Dis^{intra}_2$ of the two clusters;
    Select the maximum value $D^{intra}_{max}$ from $Dis^{intra}_1$ and $Dis^{intra}_2$;
    $Bor = \frac{Dis^{inter}}{D^{intra}_{max}}$;
    **if** $Bor > 1$ **then**
      Return $C^u = \{d_1, d_2\}$;
    **else**
      Select the cluster with the minimum intra-cluster distance from $Dis^{intra}_1$ and $Dis^{intra}_2$ as the unknown attack dataset;
      Return $C^u = \{d_1\}$;

where $c_j$ denotes the $j$-th clustering center. Since the initial clustering centers are chosen randomly, there exists the possibility that they are not real centroids. Therefore, the clustering centers need to be updated continuously, and the updating process can be represented by (14).

$$c'_j = \frac{1}{|N_j|}\sum_{s_i \in N_j} s_i \tag{14}$$

where $N_j$ is the sample point in the $j$-th class and $s_i$ is the coordinate of the sample point. Finally, the Kmeans algorithm generates $\kappa$ data clusters by continuous iterations and each cluster is an unknown attack. Since we do not know how many unknown attacks are present in the open scenario, the value of $\kappa$ is set to [2, $\kappa_{\text{max}}$]. The number of unknown attacks in the open scenario is determined based on the best clustering results (i.e., the silhouette coefficient score). Note that the reason for not choosing 1 as $\kappa$ is that the data in $D^{pca}$ is unlikely to be of one class. In addition, we consider the case that the unknown attack has only one class when $\kappa = 2$. The detailed unknown attack classification process is shown in Algorithm 2. Note that in Algorithm 2, we design a score threshold $\eta$, and determine that the clustering result is valid only if $S^{\max} > \eta$. The reason is to ensure that the unknown attack dataset $C^u$ is valid and accurate.

To improve the sample quality of the unknown attack, we need to process the obtained $C^u = \{d_1, d_2, \ldots, d_\kappa\}$, where $d$ denotes a sample cluster. It is well known that the closer sample points are to the cluster center, the lower the probability of them becoming outliers. Therefore, the basic idea of improving data quality is to select samples that are closer to the center of the cluster. Let $\mathcal{D} = \{dis_1, dis_2, \ldots, dis_i, \ldots, dis_n\}$ be the set of distances of a sample cluster, where $dis_i$ denotes the distance from the $i$-th sample to the center of the cluster. The average distance in $\mathcal{D}$ is calculated as follows:

$$dis_{avg} = \frac{\sum_{i=1}^{n} dis_i}{n} \tag{15}$$

Then, we use $dis_{avg}$ as a threshold to filter the sample points closer to the cluster center, i.e., if $dis_i < dis_{avg}$, retain this sample point, otherwise discard it. Similarly, we deal with the sample clusters in $C^u$ based on this idea to improve the sample quality of unknown attacks, and the processed $C^u = \{d'_1, d'_2, \ldots, d'_\kappa\}$. Finally, a new dataset $D^{t+1}$ is formed based on the obtained $C^u$ and the initial dataset $D^t$, and the attack recognizer is updated by utilizing $D^{t+1}$.

### *D. Update of Attack Recognizer Based on Cloud-Edge-Vehicle Collaboration*

In this subsection, we describe how the proposed attack recognizer enables auto-updating in cloud-edge-vehicle environments.

The vehicle is not suitable for performing the recognizer update task due to limited computational and storage capacity. To alleviate the computational load on vehicles, we propose a cloud-edge-vehicle collaborative model updating method, and the specific updating architecture is shown in Fig. 3. As shown in Fig. 3, the attack recognizer is deployed on the vehicle, while the unknown attack filter and unknown attack classifier are deployed on the cloud server. First, the initial attack recognizer on the vehicle forwards network flows of undeterminable type to the connected edge base station (i.e. network flows for $p^* \leq \theta$). The edge base station forwards these flows to the cloud server and forms the dataset $x^u$. Then, the cloud server inputs $x^u$ into the unknown attack filter and unknown classifier in turn and obtains the dataset $C^u$. $C^u$ contains unknown attack features present in the current scenario and the initial dataset $D^t$ contains normal traffic and known attack features. Based on datasets $D^t$ and $C^u$ we can generate a new dataset $D^{t+1}$, which contains samples and labels of normal traffic, known attacks and unknown attacks. The cloud server updates the attack recognizer using the dataset $D^{t+1}$ to ensure that the updated recognizer can identify unknown attacks present in open scenarios. It is worth mentioning that the cross-entropy loss function is used for model training to compute the losses and the Adam optimizer to update the parameters. The detailed attack recognizer update is shown in Algorithm 3.

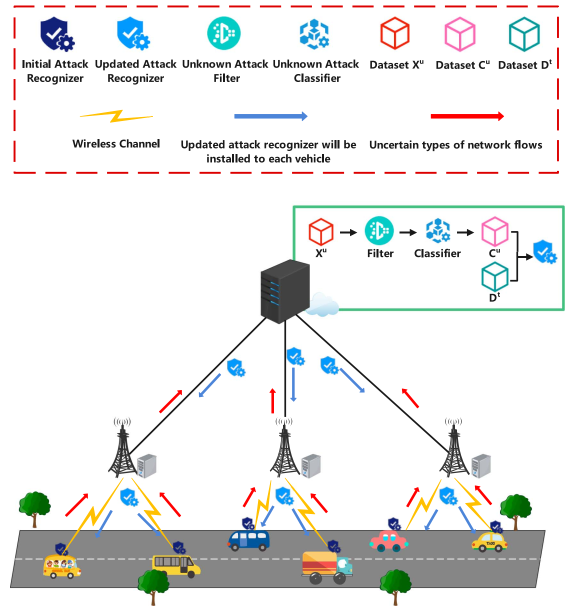


Fig. 3. A model updating method based on cloud-edge-vehicle collaboration.

**Algorithm 3:** Attack Recognizer Update.

**Input:** Dataset $D^{t+1}$

**Outuput:** New attack recognizer $F^{t+1}$

Divide the dataset $D^{t+1}$ into a training set, a validation set and a test set;
Resize the number of output SoftMax layers;
Train the new attack recognizer using the training and validation sets;
Test the new attack recognizer using the test set;
Save the model of the attack recognizer;
The cloud server sends updated model to each vehicle;
Install the updated attack recognizer in each vehicle;

## V. Experimental Results

In this section, we describe the dataset used for the experiments, the experimental settings, and the performance analysis of the proposed auto-updating intrusion detection system for vehicular network. Specifically, we will discuss five research questions.

- **Q1**: How is the overall performance of the proposed auto-updating intrusion detection system for vehicular network?
- **Q2**: Whether the proposed two-layer unknown attack filter can effectively extract the unknown attacks?
- **Q3**: Can the unknown attack classifier effectively classify unknown attacks?
- **Q4**: Whether attack recognizer based on 1D-CNN and multi-head attention mechanisms have higher accuracy compared to other algorithms?
- **Q5**: Why are vehicles not suitable for training attack recognizers?

TABLE I
DIVIDE KNOWN AND UNKNOWN ATTACKS FROM THREE DATASETS

| Dataset | Known Attacks | Unknown Attacks |
|---|---|---|
| VeReMi | DoS_Random, Const_Pos, Random_PoSoffset, Random_Speed, Delayed_Messages, Const_Speed | DoSDisruptive, TrafficSybil |
| ToN-IoT | DoS, DDoS, Injection, Scanning | XSS, Password |
| EDGE-IIoT | DDoS_ICMP, DDoS_UDP, DDoS_TCP, Vulnerability_scanner, Backdoor, DDoS_HTTP, SQL_injection, Ransomware, XSS | Uploading, Password, Port_Scanning |

Fig. 4. Baidu apollo auto-driving car.

## A. Dataset

We use open datasets VeReMi [38], ToN-IOT [39] and EDGE-IIOT [40] to train an intrusion detection system for vehicular network. The VeReMi dataset is a dataset for evaluating malicious vehicle misbehavior in vehicular network that includes DoS, false location, message replay, and other attacks. ToN-IOT and EDGE-IIOT are datasets commonly used in intrusion detection systems in recent years. The ToN-IoT dataset consists of 21,000+ network flows and contains attacks such as DoS, DDoS, XSS, etc. EDGE is a larger intrusion detection dataset that consists of millions of network flows and includes attacks such as SQL injection, port scanning, password, etc. To simulate network attacks in open scenarios, we further divide each dataset into known attacks and unknown attacks as shown in Table I. It is important to mention that the training data of the initial attack recognizer does not contain unknown attacks.

## B. Experimental Settings

*1) Experimental Environment:* The attack recognizer update task was deployed on an Xeon(R) Platinum 8336 C cloud server with 228 G of RAM and an NVIDIA A100 GPU. We use a high-performance computer as the host for the edge base station, which is equipped with an Intel(R) Core i9-11900 K CPU, 64 G of RAM, and an RTX 3090 GPU. The trained attack recognizer was deployed in the Baidu Apollo auto-driving car shown in Fig. 4. Moreover, we use the TensorFlow 2.9.0 deep learning algorithm library and the Sklearn machine learning algorithm library to build the proposed auto-updating intrusion detection system for vehicular network.

*2) Evaluation Metrics:* To evaluate the performance of the proposed vehicular network attack recognizer to identify known or unknown network attacks, we calculated the number of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN) predicted by the model. Based on these definitions, we can calculate the precision, recall, and F1 score by (16), (17), and (18), respectively.

$$Precision = \frac{TP}{TP + FP} \tag{16}$$

$$Recall = \frac{TP}{TP + FN} \tag{17}$$

$$F1 - score = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{18}$$

## C. Overall Performance of the Proposed System (Q1)

To answer Q1, we use the confusion matrix to evaluate the overall performance of the proposed system, and the results of the confusion matrix are shown in Fig. 5. The Fig. 5(a)–(c) shows the performance of the initial attack recognizer in detecting known attacks, while Fig. 5(d)–(f) shows the performance of the updated attack recognizer in detecting known attacks and unknown attacks. We can observe that the proposed vehicular network intrusion detection system can efficiently identify both known attacks at the initial time and unknown attacks after auto-updating. In addition, we note that the updated model has a reduced detection accuracy for some known attacks. The reason for this phenomenon is that a small amount of known attack data is incorrectly identified as unknown attacks. However, the updated model can still identify most of the known attacks with over 90% accuracy. It is also worth noting that the attack recognizer performs best on the TON-IOT dataset. The reason for this phenomenon is the relatively simple feature distribution of the data samples in the TON-IOT dataset. Some existing intrusion detection schemes [39], [41], [42] have also achieved good experimental results on the ToN-IoT dataset.

## D. Unknown Attack Detection Performance (Q2)

In open scenarios, constructing efficient filters for unknown attacks is the basis for reconfiguring attack recognizers. In this subsection, we will discuss the performance of the proposed two-layer filter. Fig. 6 represents the accuracy of the first layer filter in identifying normal flows and abnormal flows. Fig. 7

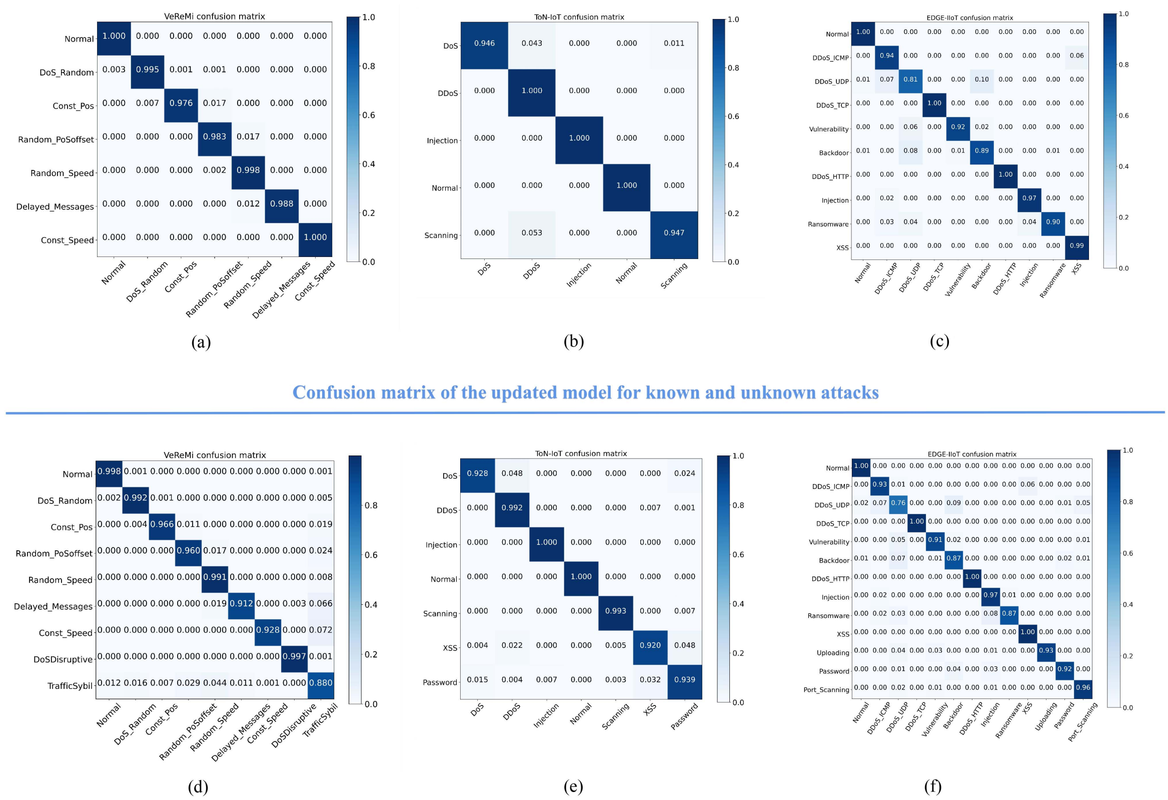


Fig. 5. Performance of the attack recognizer at initial and after update on three datasets. (a) The initial confusion matrix of VeReMi. (b) The initial confusion matrix of ToN-IoT. (c) The initial confusion matrix of EDGE-IIoT. (d) The updated confusion matrix of VeReMi (e) The updated confusion matrix of ToN-IoT. (f) The updated confusion matrix of EDGE-IIoT.

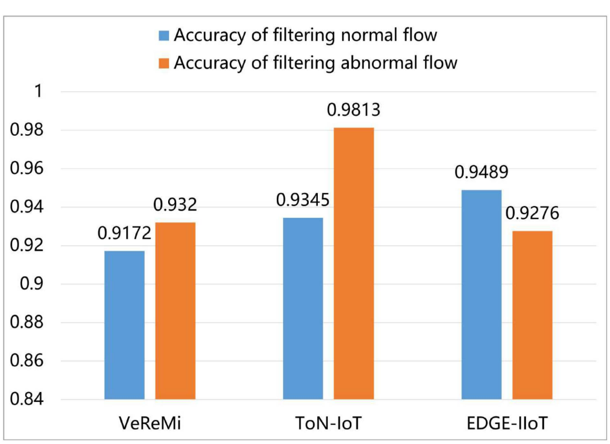


Fig. 6. Performance analysis of the first layer of filters.

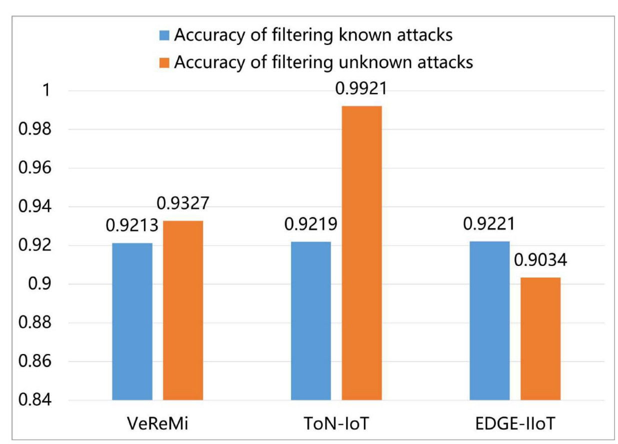


Fig. 7. Performance analysis of the second layer of filters.

represents the accuracy of the second layer filter in identifying known attacks and unknown attacks. We can observe that the first layer filter is effective in identifying anomalous flows from mixed flows, and the second layer filter is able to filter out unknown attacks from mixed attacks. In addition, the two-layer filter is more sensitive to anomalous data on ToN-IoT datasets. The reason for this phenomenon is that ToN-IoT is a smaller dataset and has low data complexity, thus the filters generated on the dataset perform better. Finally, it is important to note that we use the true labels of unknown attacks to accurately evaluate the performance of the two-layer filter.

### E. Unknown Attack Clustering Results Analysis (Q3)

To answer Q3, we analyze the clustering results of unknown attacks in this subsection. Assume that there are at maximum 10 unknown attacks in the open scenario, i.e., set $\kappa$ to 10. Then, according to the description in Algorithm 2, we perform clustering from 2 to 10 on the obtained unknown attack data and select the best clustering model by silhouette coefficients. The

TABLE II
UNKNOWN ATTACK CLUSTERING ACCURACY

| VeReMi | | ToN-IoT | | EDGE-IIoT | | |
|---|---|---|---|---|---|---|
| DoSDisruptive | TrafficSybil | XSS | Password | Uploading | Password | Port_Scanning |
| 0.98254 | 0.91579 | 0.89726 | 0.91674 | 0.90968 | 0.9163 | 0.95215 |

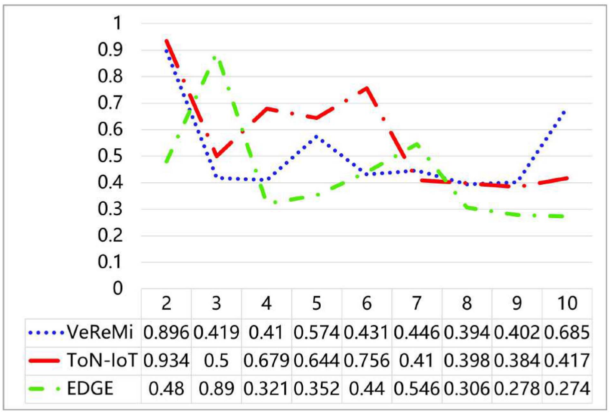


Fig. 8. Silhouette coefficient scores for the number of unknown attack types from 2 to 10.

silhouette coefficient scores for the three datasets are shown in Fig. 8. We can observe that VeReMi and ToN-IoT obtain the highest silhouette coefficients when the number of unknown attack types is set to 2, while EDGE-IIoT obtains the highest silhouette coefficients when the number of unknown attack types is set to 3. It corresponds exactly to the number of unknown attack types set in Table I.

Furthermore, we analyze the accuracy of clustering and the results are shown in Table II. We observe that the clustering accuracy for each unknown attack is higher than 89%. It should be mentioned that the higher the clustering accuracy, the higher the accuracy of the updated model to identify unknown attacks. Similarly, we used the true labels of the unknown attack data to evaluate the unknown attack classifier.

### *F. Comparison With Other Methods (Q4)*

To answer Q4, we compare the proposed attack recognizer with some existing methods [23], [43], [44] in terms of precision, recall, and F1-score. The main features of the compared schemes are as follows:

- *TLOC [43]:* To detect both in-vehicle network attacks and out-of-vehicle network attacks faced by vehicles, the scheme proposes an intrusion detection method based on five advanced CNN models such as VGG16, VGG19, Xception, Inception, and InceptionResnet, and optimizes the hyper-parameters using particle swarm.
- *LIDS [23]:* To detect the presence of false message attacks in vehicular network, an LSTM-based intrusion detection method is proposed, which utilizes the strong processing capability of LSTM on time series to detect whether a message sequence is abnormal or not.
- *NIDT [44]:* To improve the accuracy of the intrusion detection model and reduce the false alarm rate, an intrusion detection method based on CNN and BiGRU is proposed, which uses CNN to extract the spatial features of the network traffic, and then uses BiGRU to extract the time series information of the network traffic.

The comparison results are shown in Table III, we can observe that the proposed method achieves a high performance and outperforms these compared methods. The reason for this result is that 1D-CNN and the multi-head attention mechanism can effectively extract the features of network traffic and the correlation between the features. In addition, we further analyze the shortcomings of the compared methods. (1) The TLOC [43] builds an intrusion detection system by integrating five advanced CNN models, but this simple integration may result in overall performance degradation due to the poor performance of one of the base learners. (2) LIDS [23] utilizes the powerful time-series processing capability of LSTM to detect network attacks on vehicles, but LSTM suffers from gradient vanishing when processing longer sequences, resulting in poor performance. (3) Similarly, the reason for the poor performance of NIDT [44] is that BiGRU has difficulty in capturing long-term dependencies in longer sequences. Compared with these methods, the proposed attack recognizer with 1D-CNN + multi-head attention mechanism can achieve more flexible and efficient detection for network traffic. Specifically, 1D-CNN can effectively extract local features of network traffic, and multi-head attention utilizes these local features to form a global information representation, which enhances the long-range dependency between different locations and improves detection accuracy. In addition, it is important to note that our proposed intrusion detection system is composed of multiple modules, and each module works individually. This implies that the detection system is highly scalable (e.g., the attack recognizer based on 1D-CNN + multiple-attention mechanism can be flexibly replaced with an LSTM-based attack recognizer).

### *G. Comparison of Vehicle and Server Efficiency on Training Models (Q5)*

We explain why vehicles are not suitable for training attack recognizers by comparing the efficiency of vehicle and cloud server training models. The comparison results are shown in Fig. 9. We can clearly observe that as the data scale increases, the time consumed to train the model on the vehicle increases significantly, while the time consumed on the cloud server is relatively low. When the computational resources of a vehicle are occupied for a long time due to updating the attack recognizer, it is likely to affect the operation of normal safety applications,

TABLE III
COMPARISON WITH OTHER METHODS

| Methods | | VeReMi | | | ToN-IoT | | | EDGE-IIoT | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Precision | Recall | F1-score | Precision | Recall | F1-score | Precision | Recall | F1-score |
| TLOC [43] | Original | 0.86023 | 0.88494 | 0.86472 | 0.98957 | 0.98976 | 0.98938 | 0.89293 | 0.91056 | 0.90116 |
| | Updated | 0.8733 | 0.88997 | 0.87998 | 0.97411 | 0.97352 | 0.97318 | 0.90273 | 0.92023 | 0.91106 |
| LIDS [23] | Original | 0.88183 | 0.91508 | 0.8948 | 0.86693 | 0.91034 | 0.88575 | 0.7887 | 0.87617 | 0.82637 |
| | Updated | 0.88618 | 0.90176 | 0.89095 | 0.79908 | 0.83795 | 0.8155 | 0.80926 | 0.87651 | 0.83286 |
| NIDT [44] | Original | 0.91079 | 0.94504 | 0.92722 | 0.95506 | 0.95708 | 0.95351 | 0.94007 | 0.94283 | 0.94091 |
| | Updated | 0.90493 | 0.92143 | 0.9108 | 0.91885 | 0.92141 | 0.91961 | 0.89200 | 0.93169 | 0.90694 |
| Our | Original | 0.99546 | 0.99135 | 0.99339 | 0.99527 | 0.97858 | 0.98669 | 0.94634 | 0.95202 | 0.94870 |
| | Updated | 0.96212 | 0.9582 | 0.95989 | 0.92850 | 0.96744 | 0.94655 | 0.93874 | 0.94572 | 0.94118 |

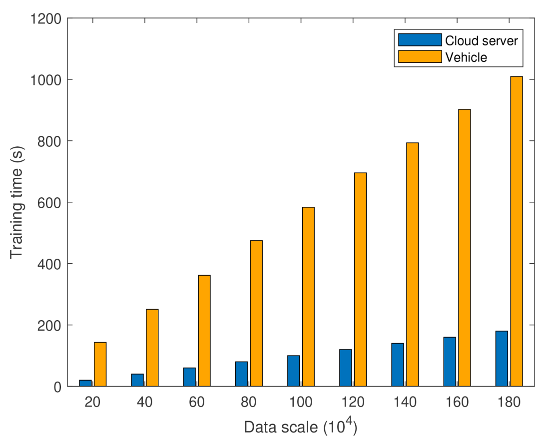


Fig. 9. Comparison of vehicle and cloud servers in terms of model training efficiency.

leading to an elevated probability of traffic accidents. In addition, the vehicle requires a large amount of data to complete the update of the attack recognizer. However, the vehicle has limited storage capacity and this training data is not important for the usual operation of the vehicle. Therefore, vehicles are not suitable for the task of updating attack recognizers in terms of both computation and storage, whereas cloud servers have more powerful computation and storage capabilities, and model updating is more efficient and independent of the surrounding environment.

## VI. CONCLUSION

In this paper, we propose an auto-updating intrusion detection system for vehicular network, which can detect known attacks and unknown attacks present in open scenarios. More specifically, the system mainly consists of an attack recognizer, an unknown attack filter, an unknown attack classifier, and an update attack recognizer. The initial attack recognizer based on 1D-CNN and multi-headed attention mechanism is able to accurately identify known attacks in open scenarios. The unknown attack filter is used to filter the unknown attack data present in the open scenario. Then, the unknown attack classifier is used to determine how many unknown attacks are present in the open scenario. Finally, the attack recognizer is updated using the known attack data and the unknown attack data. The updated attack recognizer is able to identify more network attacks. In addition, we devise a cloud-edge-vehicle model updating method that makes the vehicle not responsible for attack recognizer updating tasks, thus reducing the computational and storage load on the vehicles. Our comprehensive experiments on three open scenarios demonstrate that the proposed auto-updating vehicular network intrusion detection system can effectively identify both known and unknown attacks.

In the future, we will study a federated learning-based intrusion detection system for vehicles that enables joint training of multi-area detection models, as well as the ability to detect unknown attacks in multiple areas. Vehicular ad hoc networks are large distributed networks, and an intrusion detection system trained solely on network traffic in one region is prone to the biased flow problem, resulting in a small detection range of the model. In addition, most existing federated learning-based intrusion detection models for vehicles rarely consider the identification of unknown attacks, leading to poor robustness of the models. Therefore, building an open federated learning-based vehicular intrusion detection system is the direction of our future work.

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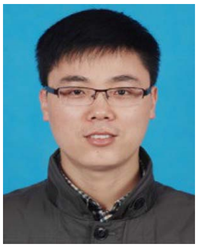

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